

BALRAM KUMAR

Data Scientist | Generative AI & LLM Engineer | AI/ML | Forward Deployed Engineer

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PROFESSIONAL SUMMARY

Data Scientist with 2+ years of experience building and shipping production ML, NLP, and Generative AI systems — including LLM-based RAG pipelines, fine-tuned language models, and real-time conversational AI. Currently working as a Junior Data Scientist, partnering directly with product and engineering teams to translate business problems into deployed AI solutions, in the style of a Forward Deployed Engineer. Proficient across the full ML lifecycle — data engineering, model development, evaluation, and cloud deployment (AWS, GCP, Azure) — with strong Python, SQL, and system-design skills. Co-inventor on 3 patents and co-author of a published book chapter on IoT security.

CORE SKILLS

Programming & Data: Python, SQL, C++, Java, Pandas, NumPy, MongoDB

Machine Learning: Scikit-learn, TensorFlow, Keras, Deep Learning, Feature Engineering, Model Evaluation, Statistical Analysis

Generative AI & NLP: LLMs, RAG, LLM Fine-Tuning, Prompt Engineering, Transformers, Semantic Search, NLTK, spaCy

Data Engineering: PostgreSQL, MySQL, RabbitMQ, Data Preprocessing, ETL Pipelines, Data Validation

Cloud & DevOps: AWS, Google Vertex AI, Azure, Docker, CI/CD, Flask, Django, Linux

Development: Node.js, REST APIs, Bash, Git

Core Strengths: Client-Facing Delivery, Problem Solving, Analytical Thinking, Communication, Leadership

PROFESSIONAL EXPERIENCE

Junior Data Scientist — ODIO

Jan 2026 – Present

- Design and deploy machine learning and Generative AI models to power data-driven product features, owning solutions from prototype through production.
- Analyze complex, large-scale datasets to extract actionable insights that guide business and engineering decisions.
- Work directly with cross-functional and client teams to scope requirements and deliver scalable AI/ML solutions on AWS and Azure.

SDE1 (Software Development Engineer I) — Frookoon Private Limited

Jan 2024 – Dec 2025

- Developed and maintained scalable backend software, improving performance and reliability across core services.
- Debugged production issues and collaborated with the engineering team on system architecture decisions.
- Applied AI, ML, and cloud technologies to improve efficiency and drive product innovation.

AI/ML Intern — TCIL-IT

Jul 2022 – Sep 2022

- Built and optimized machine learning models and data preprocessing workflows for predictive analytics projects.
- Gained hands-on experience in data analysis, Python-based model development, and data visualization techniques.

PROJECTS

Face Mask Detection System

Tech Stack: Python, OpenCV, TensorFlow, Keras, CNN

- Built a real-time face mask detection system using a custom CNN model and OpenCV to analyze live video streams, enabling automated compliance alerts and monitoring in public environments.

AI Voice Bot / Conversational AI Platform

Tech Stack: Python, Pipecat, Gemini 2.5 Flash, DeepSpeech, Cartesia TTS

- Built a real-time Generative AI voice assistant achieving sub-2-second end-to-end latency across product, finance, roadside assistance, and dealer-support workflows; implemented VAD, barge-in handling, Hindi/Hinglish-English switching, and failure recovery for reliable customer interactions.

Context-Aware Retrieval Engine (Semantic RAG)

Tech Stack: Python, FAISS, Sentence-Transformers, LangChain-style RAG

- Developed a local Retrieval-Augmented Generation (RAG) system for technical knowledge search, featuring document chunking, sentence embeddings, FAISS vector search, and query expansion; benchmarked raw vs. AI-enhanced retrieval using cosine similarity to improve relevance and performance.

EDUCATION

Bachelor of Technology, Electronics and Communication Engineering

Oct 2020 – Jun 2024

Chandigarh Engineering College, Chandigarh, India — GPA: 8.29 / 10

PUBLICATIONS

- Co-authored "Intrusion and Malware Detection in IoT," a chapter in Secure Communication in IoT (CRC Press, 2024), eBook ISBN: 9781003477327 — covering emerging technologies, challenges, and mitigation strategies in IoT security.

PATENTS

- Sleep Paralysis Detection and Prevention System — App. No. 202311038646, Issued May 6, 2023. Wearable sensor-based device using accelerometer motion detection and an ARM-based controller to safely alert individuals during sleep paralysis episodes.
- Feeding Assistive Device for Toddlers — App. No. 202311038655, Issued May 6, 2023. AI-powered assistive device using an imaging system and robotic arm for automated, accurate toddler feeding, with motion, weight, and moisture sensors.
- Hygiene Maintenance Device — App. No. 202311077323, Issued May 6, 2023. AI-powered assistive device integrating motion, weight, and moisture sensors to monitor hygiene needs and alert caregivers.